

32-bit microcontroller

How to improve ADC sampling accuracy

Application Note

Rev1.0 November 2023

Applicable objects

Product series	Product model
HC32A/F/L/M	All models

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1 Overview

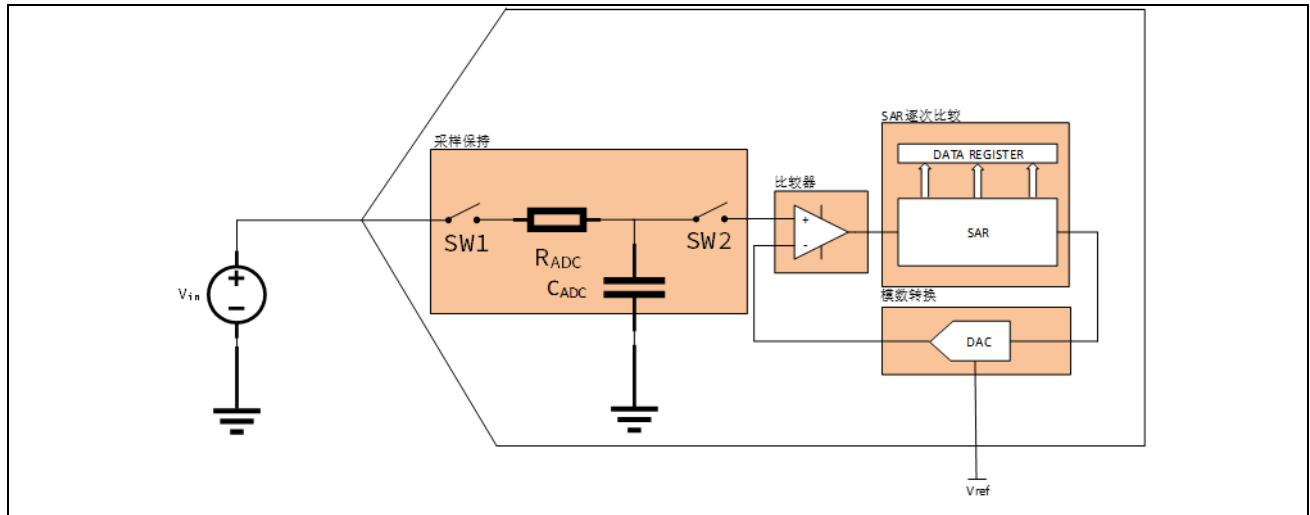
HC32A/F/L/M 系列产品是小华半导体基于 ARM cortex 内核推出的 MCU，每款 MCU 均内置 1~3 个逐次比较数模转换单元（SAR ADC）。每个单元可支持 8~16 个外部引脚输入通道或特定的内部电压、温度等信号采样。精度最高为 12 bit，最高采样转换速率可以达到 2.5 mbps，内部通道灵活映射，采样结果可通过 DMA 实现数据读取，并且能够与多功能 TIMER 联动，实现采样时序精确控制。能够满足绝大部分消费，家电及工业产品对 ADC 的需求。

本文描述 SAR ADC 的原理及固有误差、影响其采样精度的主要因素以及如何提高 ADC 的采样精度。

2 Introduction to SAR ADC

2.1 SAR ADC principle

逐次比较 ADC (SAR ADC)，顾名思义是通过输入信号与参考源电压进行逐阶比较，实现模拟信号转换为数字信号。SAR ADC 的转换过程分为两个阶段：采样阶段和转换阶段。



采样阶段：

$SW1$ 闭合， $SW2$ 断开， V_{in} 通过内部开关 R_{ADC} 对采样电容 C_{ADC} 充电。采样周期结束后进入转换阶段。

转换阶段：

$SW1$ 断开， $SW2$ 闭合，采样电容上的电压（理想情况为 V_{in} ）与 $\frac{1}{2} * V_{REF}$ 进行比较，如果 $V_{in} \geq \frac{1}{2} * V_{REF}$ ，则最高位置 1，同时 DAC 输入下一个比较值 $\frac{3}{4} * V_{REF}$ ， V_{in} 再与 $\frac{3}{4} * V_{REF}$ 比较确定次高位的值。根据比较结果再输出 DAC 值，进行比较，最后得出转换值。

2.2 SAR ADC inherent error

SAR ADC 的原理和结构决定了其存在几种固有误差。

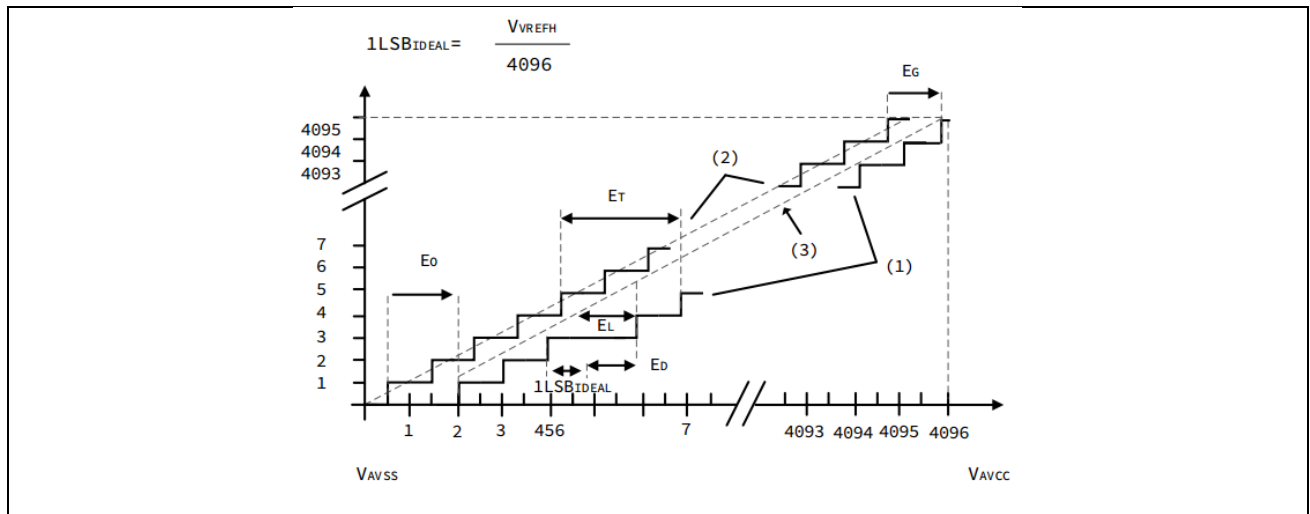


图 2-1 SAR ADC 误差

■ 偏置误差 E_0

偏置误差是指当转换值为 1 时，实际输入模拟量与理论输入模拟量的差值。偏置误差可为正，也可为负。

■ 微分非线性误差 E_D

微分非线性误差是指在转换值变化 1，实际模拟量步长与 1 LSB 之间的最大差值。

$$E_D = \text{实际模拟变化} - 1 \text{ LSB}$$

■ 积分非线性误差 E_L

积分非线性误差是指在实际模拟量与实际转换曲线之间的最大偏差值。

■ 总未调整误差 E_T

总未调整误差是指实际模拟量与理想转换曲线之间的最大误差值。

■ 增益误差 E_D

增益误差是指最后一次转换实际模拟量与理论模拟量之间的误差值。

3 ADC external error sources and solutions

3.1 Reference source

由于 SAR ADC 的原理是通过将输入信号与参考电压进行比较从而获得转换结果，即：

$$\text{Code}_{\text{ADC}} = \frac{V_{\text{in}}}{V_{\text{ref}}} * 2^N \quad (1)$$

从等式 (1) 可以看出， V_{ref} 对 ADC 的精度影响非常大，主要表现在以下方面：

3.1.1 Reference source voltage range

参考源电压需要满足：1. 高于输入信号最高电压，低于等于 ADC 电源电压。

2. 尽量接近输入信号范围。

- 参考源电压高于输入信号电压，才能准确测量到输入信号的极值。低于等于 ADC 电源电压，防止参考电压源对 ADC 电源形成电流倒灌。
- 参考源电压范围接近输入信号范围，最大程度提高理论精度。例如：

对于一个 12 bit 的 ADC，输入信号范围为 0~1.8 V。

如果参考电源电压选择为 3.3 V，则 $1 \text{ LSB} = \frac{3.3}{2^{12}} = \frac{3.3}{4096} = 0.81 \text{ mV}$ ，在这种情况下，理论上能够实现 0.81 mV 的精度。

如果参考电源电压选择为 2.0 V，则 $1 \text{ LSB} = \frac{2.0}{2^{12}} = \frac{2.0}{4096} = 0.49 \text{ mV}$ ，在这种情况下，理论上能够实现 0.49 mV 的精度。

优化建议：

- 合理的参考源电压范围可以显著提高理论采样精度，参考源略大于输入信号最大幅值。当然，在实际产品应用中，参考源不一定是任意选择，可能就是使用电源电压，那么将输入信号进行合适的放大，使其最大幅值接近参考源范围，也可以减少精度损失。对于小华半导体 MCU，参考源除了满足略大于输入信号，也需要满足规格书中的特殊要求，比如：小华半导体的 HC32F4A0 中要求。

符合	名称	最小值	典型值	最大值
V_{ref}	正参考电压	1.8 V	-	AVCC

$$0 \leq V_{\text{ref}} - \text{AVCC} \leq 1.2 \text{ V}$$

3.1.2 Reference source accuracy

由等式 (1) 不难理解, ADC 的转换值是输入信号与参考源的相对值, 参考源有较高精度, 才能保证 ADC 的输出结果精确。参考源的精度受包括温度系数, 线性调整率, 负载调整率等因素的影响。这些因素都需要关注, 尤其在在高精度场合。

假如 12-bit ADC 输入信号为 2.5 V, 参考电压 3.3 V, 则转换后的值为:

$$\frac{2.5}{3.3} * 2^{12} = 0xC1F$$

如果参考源精度为 1%, 假设为 $3.3 * 0.99 = 3.267$ V, 则转换后的值为:

$$\frac{2.5}{3.267} * 2^{12} = 0xC3E$$

$$\text{Code}_{\Delta} = 0xC3E - 0xC1F = 31 \text{ LSB}$$

所以 1% 的精度就远无法满足 1 LSB 精度要求, 需要提高精度。

优化建议:

- 在高精度的场合, 往往需要使用高精度电压基准 IC 为 ADC 提供参考源。

3.1.3 Reference source noise

在 ADC 转换过程中, ADC 内部电容切换以及充电过程会导致输入尖峰电流, 这些尖峰电流会导致参考源电压出现尖峰, 从而影响 ADC 的采样精度, 尤其当输入信号较大时, 参考源噪声影响更明显。所以减小参考源电压的噪声至关重要。

优化建议:

- 一方面可选择低噪声 LDO 或电压基准作为 ADC 参考源。
- 另一方面, 小华半导体也推荐客户在芯片参考源就近处配置低 ESR 的陶瓷电容, 进行噪声抑制, 同时为参考源提供就近储能。以 HC32F4A0 为例, 其数据手册中要求进行如下配置:

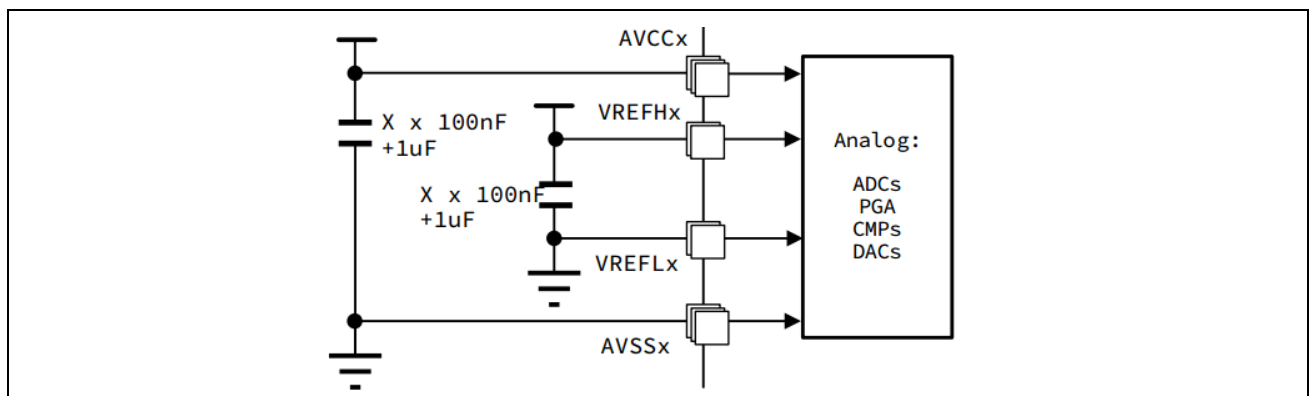


图 3-1 HC32F4A0 ADC 参考源电容

- PCB layout 时, 保证参考电源回路尽量小, 并且与数字地分割, 采用单点接地, 与模拟电源地使用同一地平面。储能和去耦电容尽量靠近电源引脚。

3.2 Power supply noise

前面讨论了参考源噪声对于 ADC 精度的影响，实际上“干净”的 ADC 供电电源对于 ADC 也是至关重要。电源噪声一方面可能影响 ADC 内部电路，也可能传导到参考源或则输入信号，导致采样误差。

DC-DC 电源：DC-DC 电源以开关形式传递能量，其拓扑中存在高频的开关信号，同时伴随对电感、输出电容充放电。开关信号有可能在电源输出或地平面感应出尖峰脉冲，而电感、输出电容充放电过程会导致输出电压有较明显纹波。这些都可以视为来自电源的噪声。

线性 LDO：线性 LDO 是基于线性环路控制，无开关信号，不会产生尖峰脉冲，且纹波较小。具备较高电源抑制比（PSRR）和低噪声的 LDO 是理想的模拟电源供电方案。

优化建议：

- 对于为模拟模块供电的电源，最好使用具有高 PSRR 及低噪声的 LDO，避免直接使用开关电源供电从而引入较大尖峰脉冲和纹波。
- 同样，小华半导体 MCU 也要求在模拟电源引脚附近配置低 ESR 的陶瓷电容进行去耦和储能。

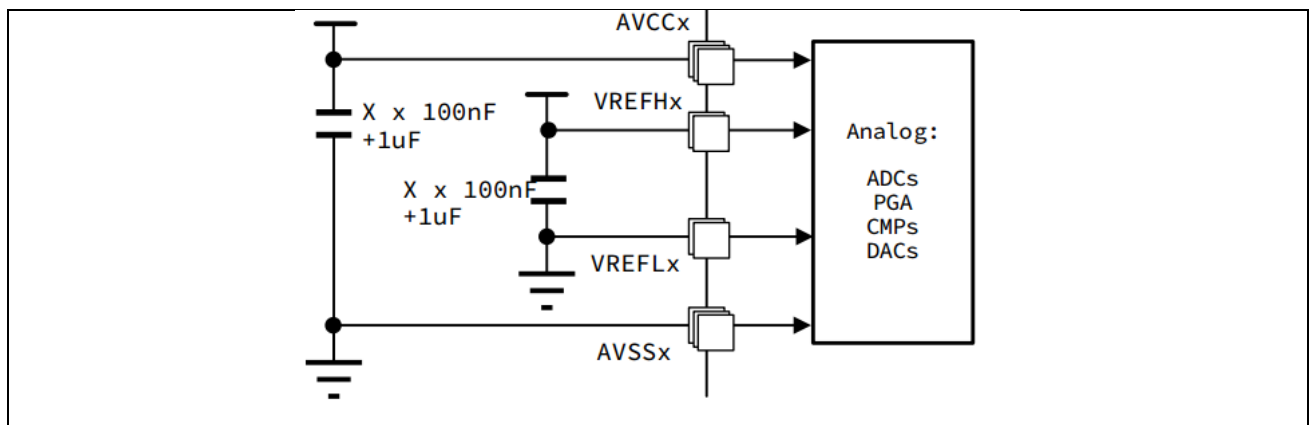


图 3-2 HC32F4A0 模拟电源电容

- PCB layout 时，保证模拟电源回路尽量小，并且与数字地分割，采用单点接地。储能和去耦电容尽量靠近电源引脚。

3.3 ADC input impedance and sampling time

在小华半导体 MCU 的规格书中 ADC 部分，一般都会列出该型号 ADC 的重要参数，如下表：

表 3-51 ADC 特性

符号	参数	条件	最小值	典型值	最大值	单位
V_{AVCC}	电源	—	1.8	—	3.6	V
$V_{REFH}^{(1)}$	正参考电压	—	1.8	—	V_{AVCC}	V
f_{ADC}	ADC 转换时钟频率	高速工作模式下 $V_{AVCC}=2.4 \sim 3.6V$	1	—	60	MHz
		低速工作模式下 $V_{AVCC}=1.8 \sim 2.4V$	1	—	30	
		超低速工作模式	1	—	8	
V_{AIN}	转换电压范围	—	V_{REFL}	—	V_{REFH}	V
R_{AIN}	外部输入阻抗	详见公式1	—	—	50	$k\Omega$
R_{ADC}	采样开关电阻	—	—	3	6	$k\Omega$
C_{ADC}	内部采样和保持电容	—	—	4	7	pF
t_D	触发器转换延迟	$f_{ADC} = 60 \text{ MHz}$	—	—	0.3	μs

其中标明了 ADC 的采样开关电阻 R_{ADC} ，ADC 的内部采样和保持电容 C_{ADC} 。从而根据公式计算允许的最大外部输入阻抗。这里提到的公式即为：

$$R_{AIN} \leq \frac{t}{C_{ADC} * \ln 2^{N+1}} * R_{ADC}$$

其中：

t：采样周期

N：ADC 位

SAR ADC 的采样电路可以简化为下图：

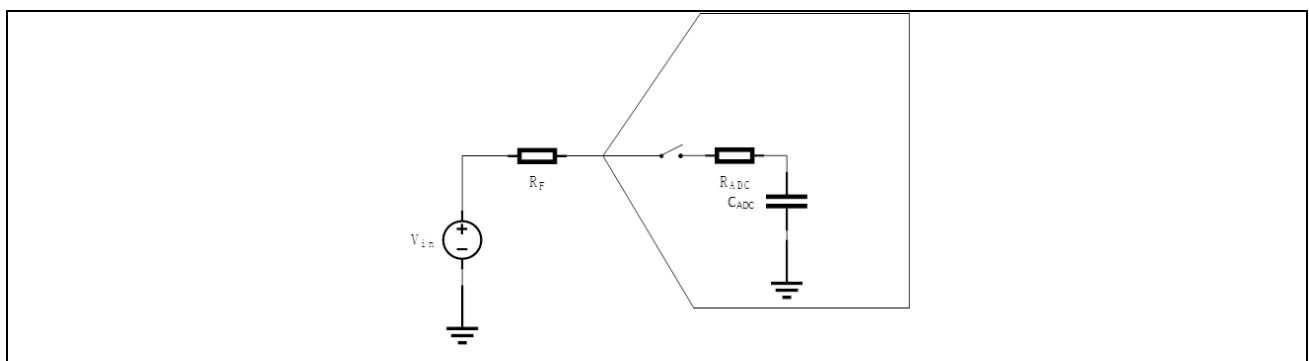


图 3-3 SAR ADC 原理框图

当 ADC 内部通道切换到某一外部引脚时，引脚上的信号电压将通过 R_{AIN} 和 R_{ADC} 对 C_S 进行充电。其充电时间常数为：

$$\tau = (R_F + R_{ADC}) * C_{ADC}$$

假设 C_S 的初始电压为 V_{t0} ，系统单位阶跃响应为：

$$V_t = V_{t0} + (V_{in} - V_{t0}) * (1 - e^{-\frac{t}{\tau}})$$

After a fixed time (i.e. sampling time t), the voltage difference between the voltage across the CADC capacitor and the signal source is:

$$V_{err} = (V_{in} - V_{t0}) * e^{-\frac{t}{\tau}} \quad (2)$$

When $V_{t0}=0$, that is, when the initial state is zero, the error is the largest. The larger V_{in} is, the larger the error is, and it is the largest when it is equal to V_{REF} .

The charging curve is as follows:

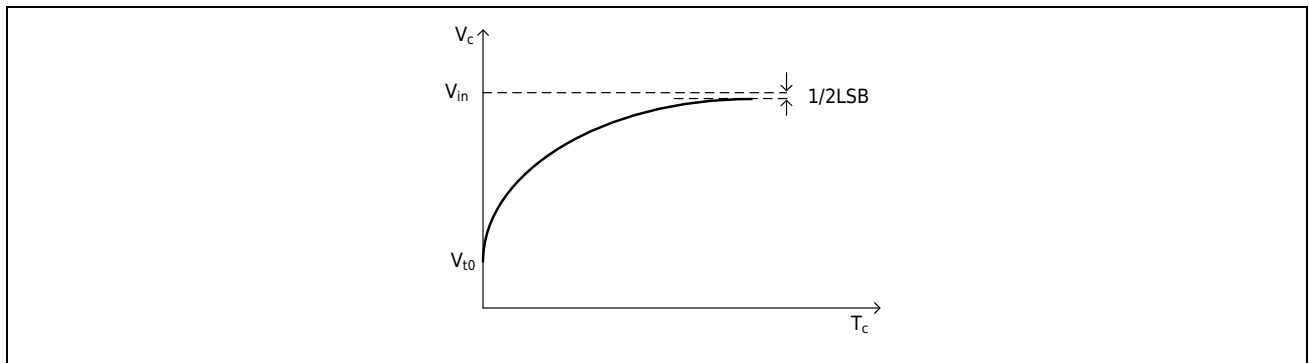


Figure 3-4 RC unit step response

If the sampling accuracy is required to reach 1/2 LSB, the error needs to meet the following requirements:

$$V_{in} * e^{-\frac{t}{(R_F + R_{ADC}) * C_{ADC}}} \leq \frac{1}{2} * \frac{V_{ref}}{2^N}$$

Simplified, we can get:

$$R_F \leq \frac{t}{C_{ADC} * \ln 2^{N+1}} - R_{ADC} \quad (3)$$

或

$$t \geq (R_F + R_{ADC}) * C_{ADC} * \ln 2^{N+1} \quad (4)$$

Therefore, in order to obtain a higher sampling rate, it is necessary to reduce the external input impedance. If the external impedance is large, it is necessary to consider adding a driver stage for impedance conversion or increasing the sampling time.

In actual applications, the ADC input pin may be connected to an external RC filter circuit, as shown in Figure 3-5.

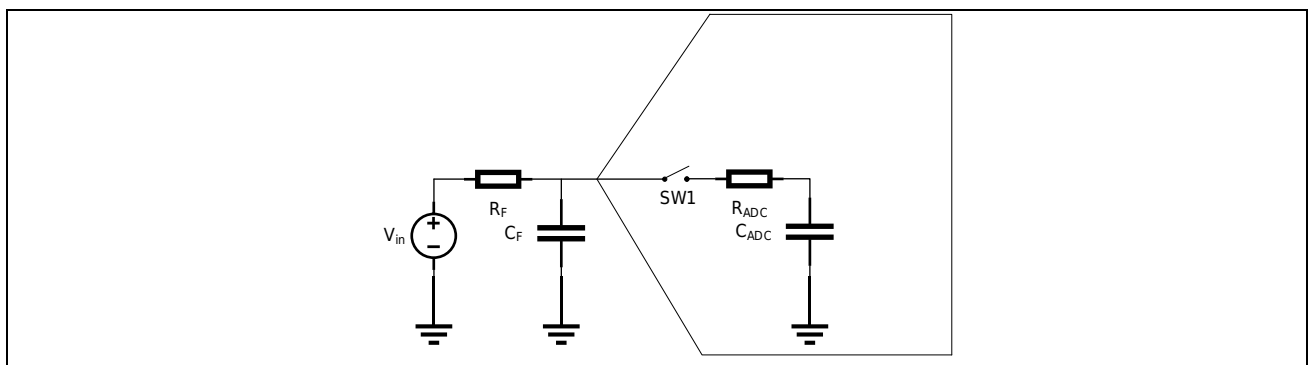


Figure 3-5 ADC external RC filter circuit diagram

In this case, charging is divided into 2 stages, and the process is shown in the figure below.

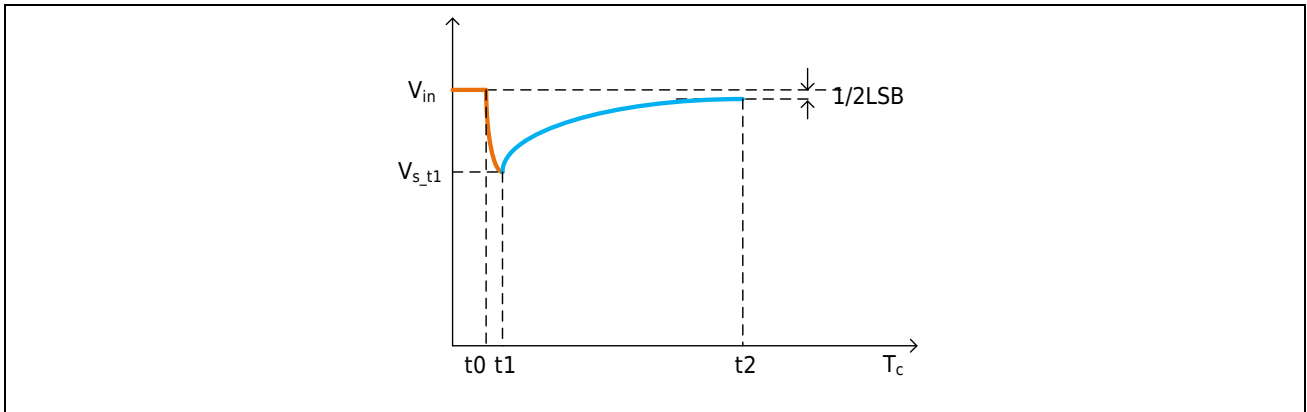


Figure 3-6 Capacitor balancing and charging process

Assume that the voltage across C_F has reached the input signal amplitude V_{in} before the ADC channel switches to the external pin. After the internal switch is closed, the charges of C_F and C_{ADC} are balanced, and finally the voltage balance is achieved.

The voltage after balance:

$$V_{s_t1} = \frac{V_{in} * C_F}{C_F + C_{ADC}} \quad (5)$$

At this equilibrium voltage, the input signal charges C_F and C_{ADC} again through $R_{AIN} + R_{ADC}$, and the starting point of charging is the voltage after equilibrium.

The time constant is:

$$\tau = (R_F + R_{ADC}) * (C_F + C_{ADC})$$

Substituting equation (5) into equation (2) yields:

$$V_{err} = \left(\frac{V_{in} * C_{ADC}}{C_F + C_{ADC}} \right) * (1 - e^{-\frac{t}{\tau}})$$

If you want to achieve 1/2LSB accuracy, you must:

$$\left(\frac{V_{in} * C_{ADC}}{C_F + C_{ADC}} \right) * \left(1 - e^{-\frac{t}{(R_F + R_{ADC}) * (C_F + C_{ADC})}} \right) \leq \frac{1}{2} * \frac{V_{ref}}{2^N}$$

After simplification, we can get:

$$R_F \leq \frac{t}{(C_F + C_{ADC}) * \ln \frac{C_{ADC}}{C_F + C_{ADC}} 2^{N+1}} - R_{ADC} \quad (6)$$

When $C_F = 0$, equation (6) is consistent with equation (3).

Here, the values of R_F and C_F also need to consider their bandwidth. Usually, in order to ensure that the input signal is not distorted after passing through the filter, the following conditions are met:

$$\tau_{AIN} \geq 10 * \tau_{FILT}$$

Right now:

$$f_{FILT} \geq 10 * f_{AIN} \quad (7)$$

The 3dB bandwidth of the RC filter circuit is:

$$f_{3db} = \frac{1}{2\pi} f_{FILT} \quad (8)$$

In addition, in actual systems, the initial voltage of Cs is often the voltage of the previous channel, so if the input voltages of two consecutively sampled channels differ greatly, the sampling accuracy of the first channel on the second channel may be affected.

Optimization suggestions:

- Ensure sufficient sampling time.
- Reduce external input impedance. For high-impedance input sources, it is necessary to increase the drive level.
- Add RC filtering to suppress input signal noise, and the RC bandwidth needs to be considered.
- Avoid arranging two signals with large voltage differences to adjacent sampling channels as much as possible.

3.4 Input signal

According to equation (1), it is not difficult to find that in addition to the reference source, another factor directly related to the conversion result is the input signal itself. When the input signal is small, the input signal noise has a significant impact.

If the impedance is too large, resulting in insufficient output current capacity, the sampling capacitor cannot be fully charged during high-speed sampling.

For AC signals, it is also necessary to pay attention to whether the signal has spike pulses or ripples. If an amplifier or RC filter circuit is used at the front end of the ADC. The bandwidth of the amplifier and the bandwidth of the RC need to ensure that the input signal is not distorted.

Optimization suggestions:

- Reduce the signal source impedance to provide sufficient charging current.
- Add a suitable RC filter to reduce the input signal noise. The RC bandwidth can be roughly estimated according to equations (7) and (8), or it can be accurately calculated according to the following formula:

$$\tau_{FILT_MAX} = \frac{t}{\ln \left(\frac{2^{N+1} * 2\pi f_{AIN} * V_{peak} * t_{CONV} * C_{ADC}}{V_{REF} * (C_F + C_{ADC})} \right)}$$

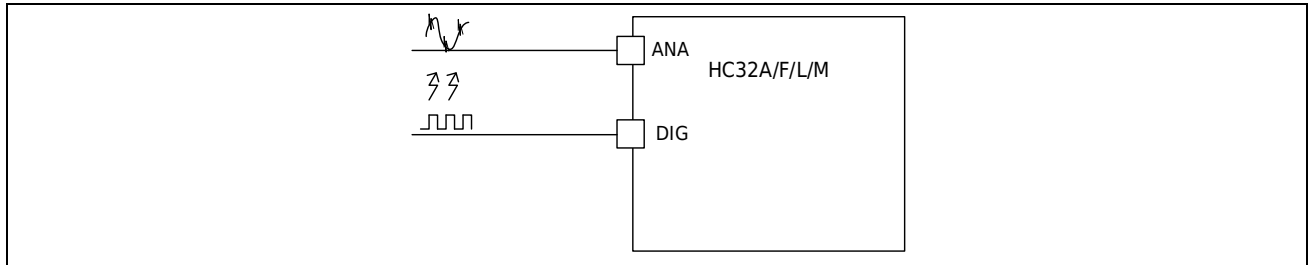
The 3 dB bandwidth of the RC filter circuit is:

$$f_{3dB} = \frac{1}{2\pi * \tau_{FILT_MAX}}$$

- Add an amplifier for impedance conversion.
- Trigger ADC sampling through accurate timing to avoid noise points of the input signal.

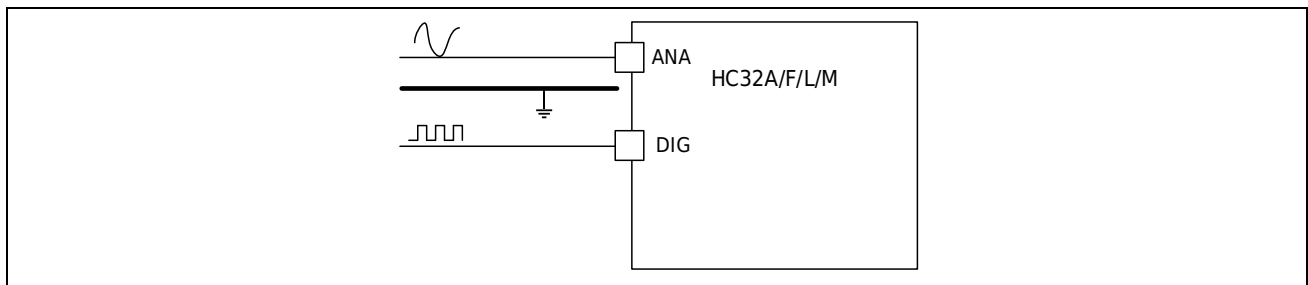
3.5 IO crosstalk

IO crosstalk may come from interference from adjacent lines on the circuit board, or from coupling inside the MCU port. This is especially true when there is a continuous high-speed square wave signal (communication signal, PWM signal, etc.) in close proximity to the ADC line. This square wave signal will generate spikes on the adjacent ADC line through capacitive coupling, affecting the ADC sampling accuracy.



Optimization suggestions:

- When arranging pin functions, try to avoid PWM, high-speed communication ports and ADC pins being adjacent.
- In PCB layout, keep ADC input signal lines away from PWM or high-speed communication lines.
- Use ground wires to isolate ADC input signal lines from PWM or high-speed communication lines.



3.6 Improving accuracy by software

Improving ADC sampling accuracy by optimizing power supply noise and external circuits is a fundamental means. However, in actual applications, the system electrical environment is often very complex, hardware improvements will also increase costs to a certain extent, and ADC inherent errors and white noise cannot be improved by hardware. Therefore, improving sampling accuracy by software is also an important means.

Optimization suggestions:

- Software filtering, according to noise characteristics, real-time, software overhead and other factors, use appropriate software filtering methods, such as average filtering, median filtering, first-order filtering, IIR and other filtering methods.
- Oversampling, oversampling can effectively improve the SNR of ADC and increase the effective bits of ADC data. It is suitable for occasions where the requirements for time real-time are not high and the MCU storage resources are relatively rich.

4 Summary

Successive approximation ADC (SAR ADC) has the advantages of fast speed, high accuracy and low cost, and is widely used in various MCUs. This application note briefly describes the working principle and inherent error of SAR ADC. It focuses on the factors that have a great impact on ADC sampling accuracy and how to optimize the design in practical applications. For example: reducing system noise, controlling input impedance, extending sampling time, etc. In addition, in some applications, it is also a common way to reasonably use other peripherals to trigger ADC sampling to avoid noise.

In short, ADC, as the last link in the signal chain, is crucial to the stability, accuracy and real-time performance of control. In addition to being related to the ADC itself, the power supply and external circuit design also need to match the ADC. Xiaohua Semiconductor is committed to providing high-performance MCUs, and also provides customers with strong technical support to reduce the burden on R&D personnel and shorten the time to market.

Revision history

Version	Date	Revision Content
Rev1.0	2023/11/08	Initial version released.